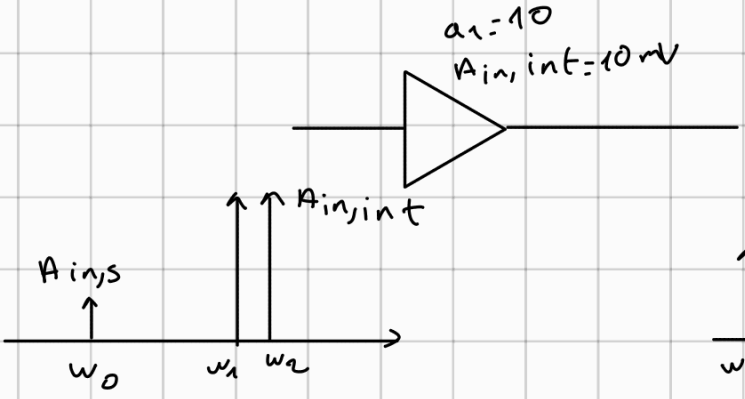
Ders 8

↓ Ders 6 'dakisorn

Ex:



$$IM3 = \frac{\frac{3}{4} a_3 A_{int}^3}{a_1 A_{int}} = \frac{A_o(IM3)}{A_{o,int}}$$

$$A_{IIP3}^2 = \frac{A_{out}(w_1,w_2)}{A_{out}(IM3)} A_{int}^2$$

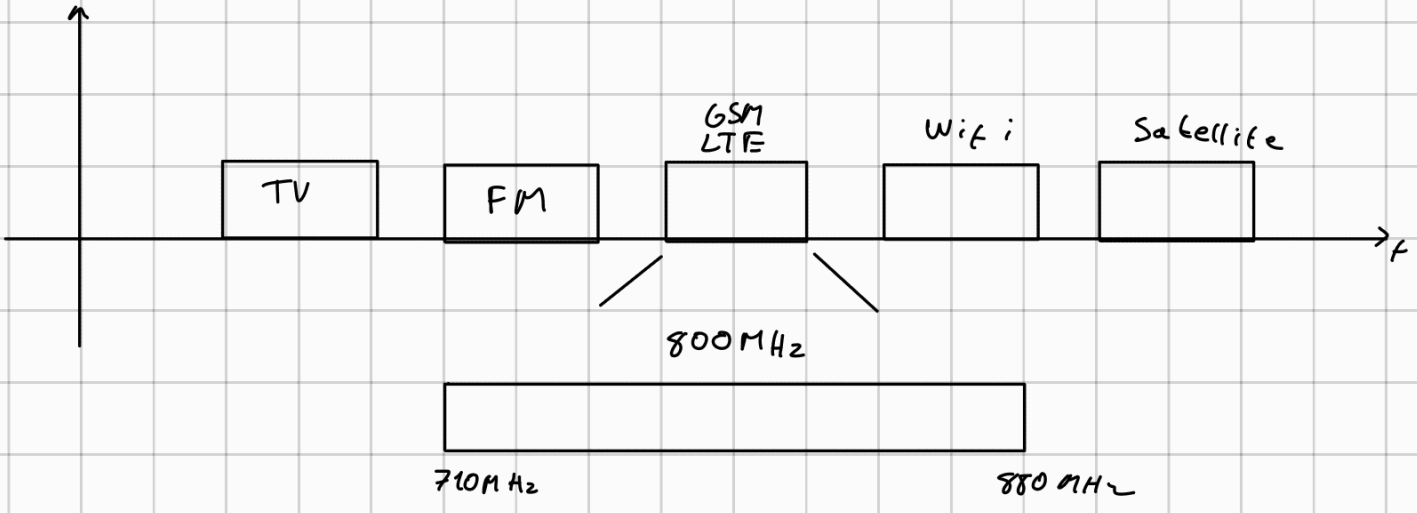
$$\Rightarrow A_{out}(IM3) = \frac{A_{out}(w_1,w_2) A_{in,int}^2}{A_{IIP3}^2} = \frac{a_1 A_{in,int}^3}{A_{IIP3}^2}$$

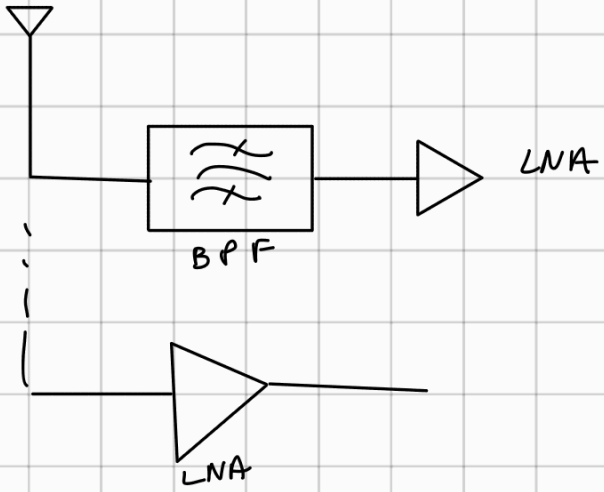
($a_1 A_{in,int}$; $A_{in,int} \rightarrow 10mV$)

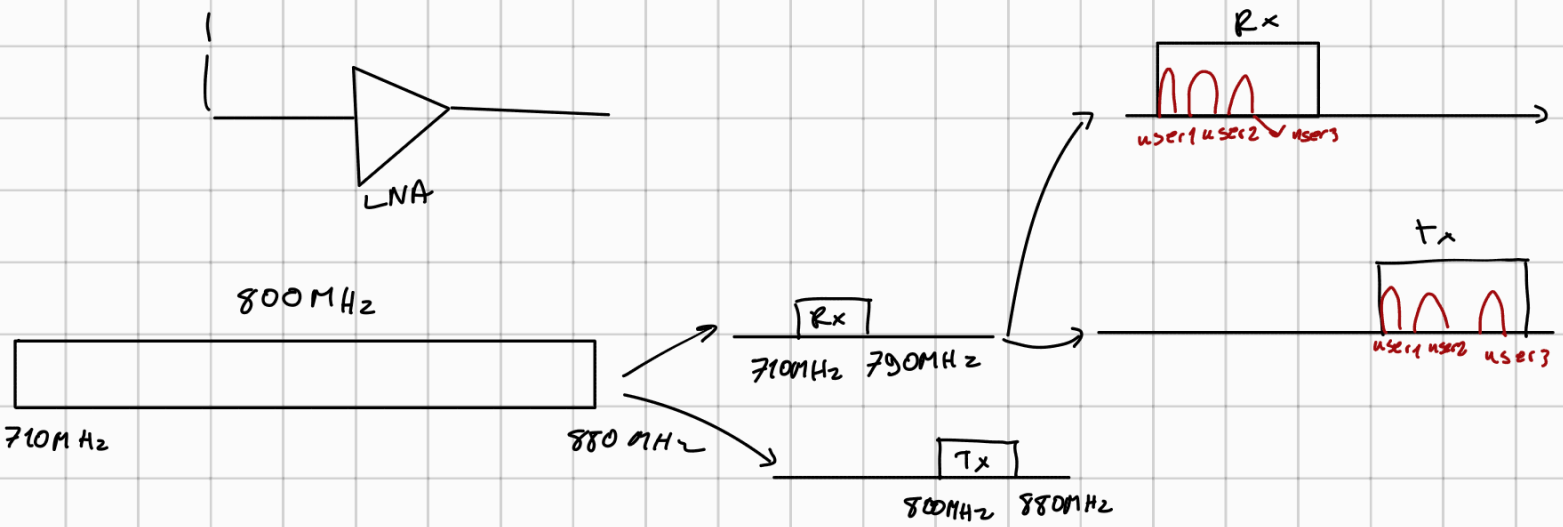
$$A_{out}(IM3) = 1,56 \cdot 10^{-3} V$$

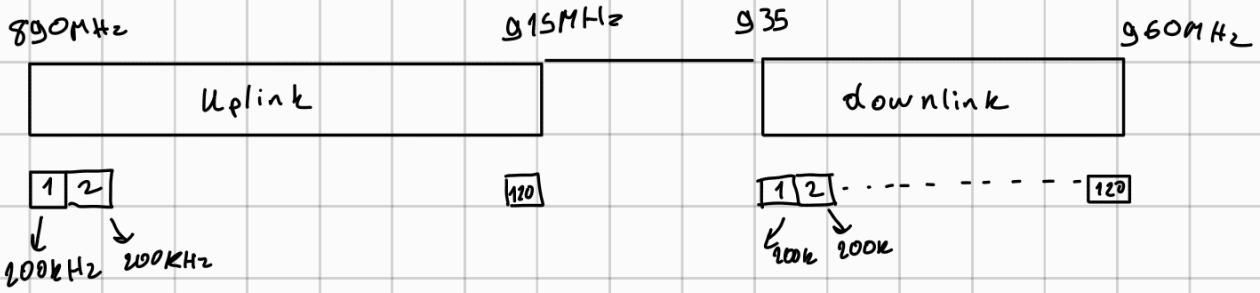
$$IM3 = \frac{1,56 \cdot 10^{-3}}{a_1 \cdot A_{in,int} \rightarrow 10 \cdot 10 \cdot 10^{-3}} = 0,0156$$

Modulation Concepts:





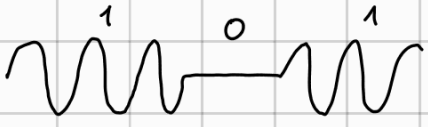






Modulation Concepts

1-Detectability

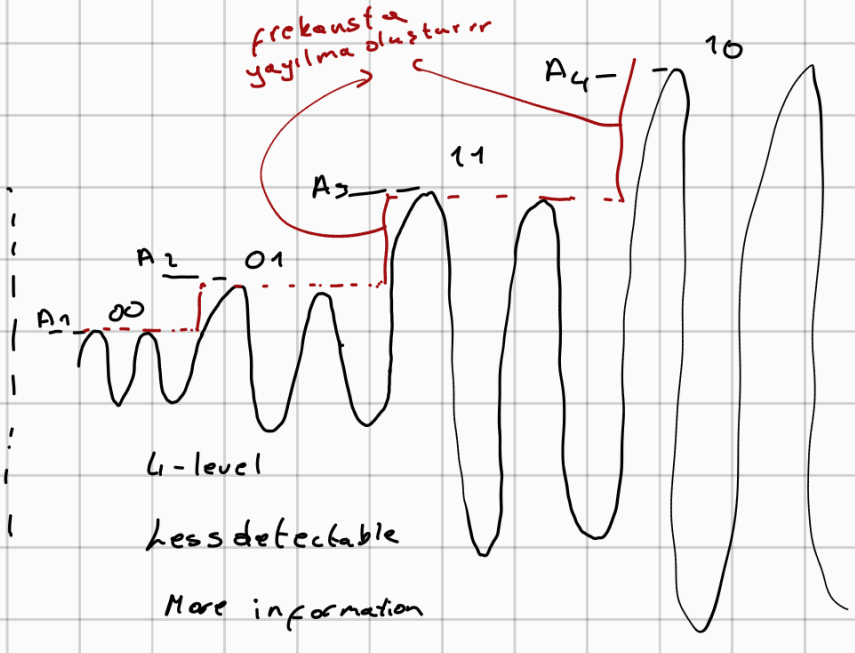


2-level

More dectable

Less information

Less susceptible to noise



4-level

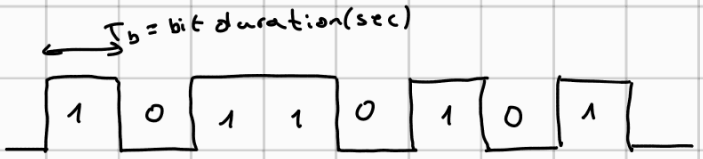
Less detectable

More information

More susceptible to noise

2) Bandwidth Efficiency

$$\eta = \frac{\text{transmission bitrate (bps)}}{\text{transmission BW (Hz)}} = \frac{\frac{bit}{sec}}{Hz}$$

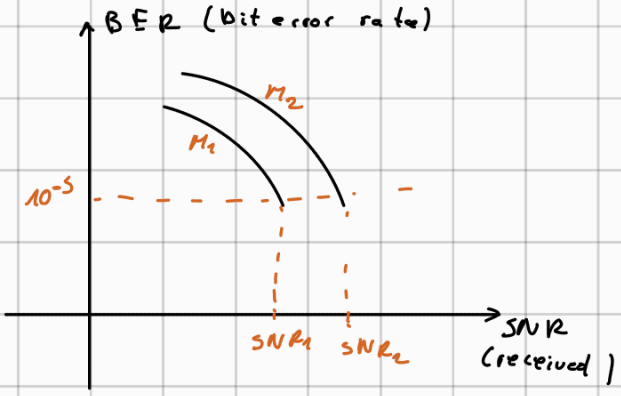


$$R_b = \frac{1}{T_b} = \text{data rate } (1/\text{sec}) \text{ bps}$$

3) Max (channel capacity (Shannon)

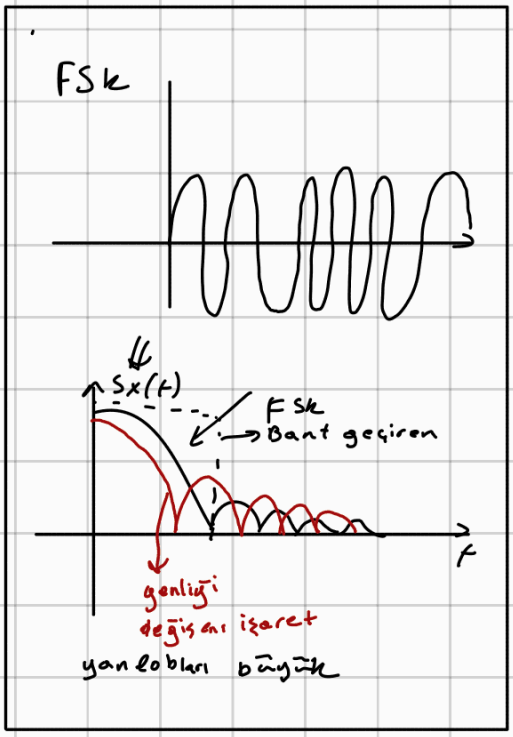
$$C = \underset{(\omega)}{B} \cdot \log_2 (1 + \underbrace{SNR}_{\text{received}})$$

4) Power Efficiency



$SNR_1 < SNR_2$

M_1 power efficient



5) Peak-to-Average Ratio
(tepe gücünün ortalama güce oranı)

$$PAPR = \frac{\text{Power of signal when carrier is at peak}}{\text{average power of the carrier}}$$

(peak → maksimum güç)

$PAPR_{AM} = 4,26dB$

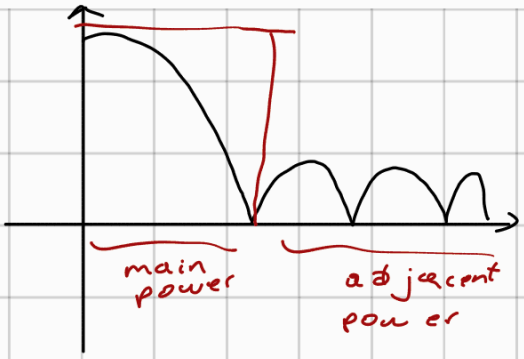
$PAPR_{FM} = 3dB$

6) Adjacent channel interference (ACI)
Adjacent channel Power Ratio (ACPR)

→ komşu kanala girişim oranı

$$ACPR = \frac{\text{Power in adjacent channel}}{\text{Power in main channel}}$$

Ne kadar küçükse o kadar iyi

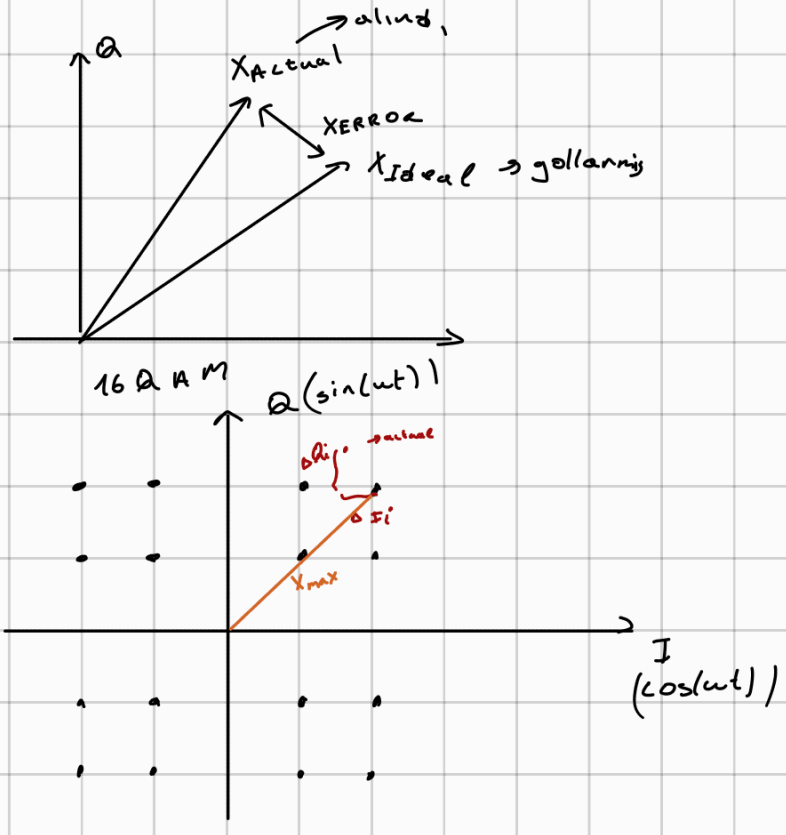


7) Implementatio Margin

Theoterically to achive any BER required SNR = A

'practically " " A . k → implementing Margin

8) Error vector Magnitude (EVM)



$$EVM = \sqrt{\frac{\frac{1}{N}\sum_{i=1}^{N}(\Delta I_i^2 + \Delta Q_i^2)}{X_{max}^2}}$$

9) Modulation ERRor rate (MER)

$$MER = \frac{\frac{1}{N}\sum_{i=1}^{N}(I_i^2 + Q_i^2)}{\frac{1}{N}\sum_{i=1}^{N}(\Delta I_i^2 + \Delta Q_i^2)}$$

$$= \frac{\text{Average signal power}}{\text{Average error power}}$$